

Medical-Supply ERP

User Manual

Sudan MedSupply Co.

Built on ePHEM / Odoo 18 Community

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1. Introduction

This manual describes the **Medical-Supply ERP** — an integrated business system for a company that imports, stores, and distributes medical supplies and pharmaceuticals. The system covers the full operational chain: master-data configuration, inventory and warehousing, procurement, sales, and accounting, with native **multi-currency** support (Sudanese Pound as the base currency and US Dollar as a reference currency).

The system is built on **Odoo 18 Community** (the ePHEM deployment) and uses only Community and open-source (OCA / Odoo-Mates) modules — no proprietary Enterprise licences are required.

1.1 Who this manual is for

- **Administrators** who configure master data — warehouses, items, partners, banks, currencies and exchange rates.
- **Procurement officers** who raise purchase orders and receive goods.
- **Sales staff** who quote, sell, and deliver to customers.
- **Accountants** who manage invoices, payments, bank reconciliation and reports.

1.2 Key capabilities at a glance

Area	What you can do
Configuration	Define company, currencies & rates, customers, suppliers, banks, cash safes, units, warehouses and product categories.
Inventory	Register items with lot/batch and expiry tracking; manage on-hand stock and valuation.
Warehouse	Run multiple warehouses and internal locations (e.g. cold storage, quarantine); receipts, deliveries and internal transfers.
Procurement	RFQ → Purchase Order → Receipt → Vendor Bill, with vendor price lists and multi-currency purchasing.
Sales	Quotation → Sales Order → Delivery → Customer Invoice, with price lists per currency.
Accounting	Customer invoices, vendor bills, payments, bank reconciliation, financial reports, fiscal year and dunning.
Multi-currency	SDG base + USD reference; dated exchange-rate history that is easy to update.

2. System Overview

2.1 Building blocks

The ERP is organised into cooperating Odoo applications. Each application adds menus, documents and reports, but they share the same products, partners and accounting.

Application	Purpose
Inventory	Items, stock, warehouses, locations, lots, expiry, transfers.
Purchase	Supplier orders and goods receipts.
Sales	Customer quotations, orders and deliveries.
Invoicing / Accounting	Invoices, bills, payments, journals, reports.
Contacts	Customers and suppliers (a single contact can be both).
Settings	Company, currencies, users, and feature switches.

2.2 Logging in

1. Open the system URL in a web browser (for the local demo: **http://localhost:8069**).
2. If a database list appears, choose **erpmedsupply**.
3. Enter your user name and password. The default administrator is **admin / admin** on the demo database — change this password before going live.

IMPORTANT — Change default credentials

The demo administrator password (**admin**) and the database master password must be changed before the system is used with real data.

2.3 Finding your way around

- The **top menu bar** switches between applications (Inventory, Purchase, Sales, Accounting...).
- Each application has its own **Configuration** sub-menu for setup tasks.
- Lists can be filtered, grouped and exported; most documents follow a **draft → confirmed → done** lifecycle shown by status buttons in the top-right.

3. Configuration (Central Admin)

Configuration is the heart of the system. Complete these steps **in order** before recording day-to-day transactions, because some settings (especially the base currency and chart of accounts) are difficult to change once transactions exist.

3.1 Company

1. Go to **Settings → Users & Companies → Companies** and open your company.
2. Set the legal name, address, tax number and logo.
3. Set the **Country** to Sudan.

3.2 Currencies and exchange rates

The system uses **SDG (Sudanese Pound)** as the company base currency and **USD** as a reference currency for imports and price quotations.

1. Enable multi-currency: **Settings → Accounting → Currencies → tick Multi-Currencies**.
2. Activate the currencies you use: **Settings → Currencies**, switch on **SDG** and **USD**.
3. Confirm the company base currency is **SDG** (set on the company / chart of accounts).

Recording a new USD exchange rate

Exchange rates are stored as **dated records**, so the full history is preserved and any past document is valued at the rate that applied on its date.

1. Open **Accounting → Configuration → Currencies** and click **USD**.
2. Go to the **Rates** tab and click **Add a line**.
3. Enter the **date** and the rate. The system lets you type the intuitive figure — **how many SDG equal 1 USD** (e.g. 700).
4. Save. The new rate applies to all documents dated on or after that day.

TIP — Easy rate updates

Because each change is a new dated line, you simply add a line whenever the rate moves — you never overwrite history. Reports and documents automatically use the correct rate for their date.

NOTE — Automatic rates (optional)

Community Odoo does not fetch rates automatically. If you want scheduled updates from a provider, the OCA **currency_rate_update** module can be added later; otherwise rates are maintained by hand as above.

3.3 Customers and suppliers

Customers and suppliers are both **Contacts**. A single contact can be both a customer and a supplier — do not create duplicates.

1. Open **Contacts → New**.
2. Choose **Company** or **Individual** and enter name, address and phone.
3. On the **Sales & Purchasing** tab set the price list, payment terms and (for foreign suppliers) the purchase currency.
4. On the **Accounting** tab set receivable/payable accounts and add bank accounts if needed.

Field	Used for
Customer / Vendor	Determines whether the contact appears in Sales or Purchase.
Purchase currency	Default currency for that supplier's orders (e.g. USD for imports).
Price list	Which selling prices and currency apply to this customer.
Bank accounts	Needed to register or batch payments.

3.4 Banks and money safes

Money flows are tracked through **journals**. A **bank** needs both a bank record and a bank journal; a **money safe / cash box** is a **cash journal**.

1. Create the institution: **Accounting → Configuration → Banks → New** (name, SWIFT/BIC).
2. Create a **Bank** journal: **Accounting → Configuration → Journals → New**, Type = **Bank**. Set a currency for a foreign-currency account (e.g. a USD bank account).

3. Create a **Cash** journal for each physical safe: Type = **Cash** (e.g. *Main Cash Safe (SDG)*, *USD Cash Safe*).

NOTE — Why a cash journal is a 'safe'

There is no separate 'safe' object. Each physical cash box or till is represented by its own **Cash journal**, so its balance and movements are tracked and reconciled independently.

3.5 Units of measure

Enable **Units of Measure** (Settings) to buy, stock and sell in different units — for example purchase a *Box of 100* but issue *units*. Define units and their conversion factors under **Inventory → Configuration → Units of Measure**.

3.6 Warehouses and locations

- 1. Go to **Inventory → Configuration → Warehouses** and create one per physical site (name + short code, e.g. *Khartoum Central / KRT*).
- 2. Set the number of receipt and delivery steps on each warehouse as needed.
- 3. Create internal **Locations** under a warehouse for special storage: **Cold Storage (2-8°C), Quarantine, Expired / Damaged**.

TIP — Cold chain

Temperature-controlled storage is modelled as an internal **location**. A **putaway rule** can automatically route a product (e.g. insulin) into Cold Storage when it is received.

3.7 Product categories

Product categories control how stock is costed and removed.

Setting	Recommended value	Effect
Costing Method	FIFO	Stock is valued first-in-first-out.
Inventory Valuation	Manual / Periodic*	Avoids needing valuation accounts up front.
Force Removal Strategy	FEFO	Goods nearest to expiry are issued first.

NOTE — Automated valuation

*The demo uses periodic valuation so goods movements post without extra account setup. To post stock value to the ledger in real time, set the category's Stock Input/Output/Valuation accounts and a Stock journal, then switch Valuation to **Automated**.

4. Inventory & Items

4.1 Registering an item

1. Open **Inventory** → **Products** → **Products** → **New**.
2. Enter the name and **Internal Reference** (SKU) and **Barcode**.
3. Set **Product Type** = **Goods** and switch on **Track Inventory** so stock is counted.
4. Assign the **Product Category** (e.g. Pharmaceuticals).
5. Set the **Cost** and **Sales Price**.
6. On the **Inventory** tab set **Tracking** = **By Lots** and enable **Expiration Date** for medicines and consumables.

4.2 Lot / batch and expiry tracking

Lot tracking records which **batch** each unit belongs to; expiry tracking records the **expiration date** of each batch. Together they give full traceability and drive FEFO.

- On receipt the system asks for the **lot number** and **expiry date**.
- Stock can be traced by lot for recalls and audits.
- Near-expiry stock is highlighted and issued first under FEFO.

4.3 Checking stock on hand

Open a product and use the **On Hand** / **Forecasted** smart buttons, or go to **Inventory** → **Products** → **Products** and review the on-hand column. **Inventory** → **Reporting** provides valuation and movement analysis.

5. Warehouse Operations

5.1 Receipts (goods in)

1. Receipts are generated from confirmed purchase orders (see Procurement).
2. Open the receipt, enter the **lot number** and **expiry date** for each line, then **Validate**.
3. Stock is added to the destination location; putaway rules may route it (e.g. to Cold Storage).

5.2 Deliveries (goods out)

1. Deliveries are generated from confirmed sales orders.
2. The system reserves stock — under **FEFO** it selects the soonest-to-expire lots.
3. **Validate** the delivery to ship the goods and reduce stock.

5.3 Internal transfers

Move stock between locations or warehouses — for example from the main store to **Cold Storage**. Use **Inventory → Operations → Transfers**, choose the source and destination locations, add the product and lot, and validate.

5.4 Reordering rules

Set minimum/maximum levels on critical items (**Reordering Rules**). When stock falls below the minimum the system proposes a purchase (for *Buy* products) to top up to the maximum, preventing stock-outs of essential supplies.

6. Procurement

6.1 The purchasing flow

1. **Purchase → Orders → New** creates a Request for Quotation (RFQ).
2. Select the supplier; add products and quantities. Prices come from the vendor price list.
3. **Confirm Order** turns the RFQ into a Purchase Order and creates a **Receipt**.
4. Receive the goods (entering lots and expiry).
5. **Create Bill** generates the vendor bill; post it and register payment.

6.2 Vendor price lists

On a product's **Purchase** tab, add one line per supplier with the price, currency, minimum quantity and lead time. Import suppliers can be priced in **USD**.

6.3 Multi-currency purchasing

A purchase order can be issued in the supplier's currency (e.g. USD). The amount is converted to SDG using the exchange rate on the order/bill date, so the accounting value is always correct in the base currency.

TIP — Bill control

Set **Bill Control = Received quantities** so you only pay for what actually arrived — important for medical goods.

7. Sales

7.1 The selling flow

1. **Sales** → **Orders** → **New** creates a quotation.
2. Select the customer; the customer's price list sets prices and currency.
3. Add products and quantities and send or confirm the quotation.
4. **Confirm** creates the Sales Order and a **Delivery**.
5. Validate the delivery (FEFO selects the lots), then **Create Invoice** and post it.

7.2 Price lists and currency

Each price list has a currency. Domestic sales use an **SDG** price list. The sales order's currency follows its price list, so assign the correct price list to each customer.

IMPORTANT — Currency comes from the price list

If invoices appear in the wrong currency, check the price list assigned to the customer — the order currency follows the price list, not the contact's country.

8. Accounting

8.1 What is included

Community Odoo provides **Invoicing**; the bundled OCA / Odoo-Mates modules extend it into a full **Accounting** system.

Capability	Provided by
Customer invoices, vendor bills, payments, taxes	Invoicing (core)
Full Accounting menu, asset, budget, fiscal year, recurring	om_account_accountant suite
Financial reports (Balance Sheet, P&L, ledgers, tax, aged)	accounting_pdf_reports
Cash Book / Day Book / Bank Book	om_account_daily_reports
Bank reconciliation (interactive matching)	account_reconcile_oca
Customer follow-up / dunning	om_account_followup

8.2 Customer invoices and vendor bills

- Customer invoices are normally created from sales orders and posted from the Accounting app.
- Vendor bills are created from purchase orders / receipts and posted before payment.
- Each posted document generates the matching journal entries automatically.

8.3 Payments and the money safe

Register a payment from an invoice or bill and choose the journal — a **bank** account or a **cash safe**. Reconcile cash safes daily using the **Cash Book / Day Book** reports.

8.4 Bank reconciliation

Use **Accounting** → **Bank** to match bank-statement lines against invoices and payments with the interactive reconciliation screen (account_reconcile_oca).

8.5 Reports

- **Balance Sheet** and **Profit & Loss** for financial position and performance.
- **General / Partner Ledger, Trial Balance, Aged Receivable / Payable.**
- **Tax Report** for filing; **Day/Cash/Bank Book** for daily cash control.

8.6 Fiscal year and follow-up

- Define the **Fiscal Year** and set **lock dates** after each period close.
- Configure **Follow-up levels** (e.g. 15 / 30 / 60 days) to chase overdue customers.

9. Multi-Currency Operations

The system keeps accounts in **SDG** while letting you transact in **USD**.

- **Base currency:** SDG — all ledgers and reports are in SDG.
- **Reference currency:** USD — used for import purchases and quotations.
- **Rates:** stored as dated lines; the rate on a document's date is used for conversion.
- **Foreign journals:** a USD bank or cash journal holds balances in USD and is revalued to SDG at period close.

Document	Currency source
Purchase order	Supplier's purchase currency / vendor price list
Vendor bill	Inherited from the purchase order
Sales order	Customer's price list
Customer invoice	Inherited from the sales order

10. Medical-Supply Specifics

- **Batch traceability** — every medicine is tracked by lot for recalls and audits.
- **Expiry control** — expiration dates are captured per lot and stock nearing expiry is flagged.
- **FEFO** — deliveries automatically issue the earliest-expiring stock first.
- **Cold chain** — temperature-sensitive items (e.g. insulin) are stored in a Cold Storage location, with putaway rules routing them there on receipt.
- **Quarantine / damaged** — separate locations isolate non-sellable stock.
- **Imports** — purchases in USD are valued in SDG at the current rate; landed costs (freight, customs) can be folded into unit cost.

11. Demo Data Reference

The demonstration database **erpmedsupply** is pre-loaded with realistic data so every feature can be explored immediately.

Item	Details
Company	Sudan MedSupply Co. (base currency SDG)
Currencies	SDG (base) + USD (reference), with 5 dated USD rates (600 → 700 SDG)
Warehouses	Khartoum Central (KRT), Port Sudan (PRT)
Locations	Cold Storage (2-8°C), Quarantine, Expired/Damaged
Products	10 medical items with lot & expiry tracking
Customers	Khartoum Teaching Hospital, Omdurman Family Clinic, Sudanese Red Crescent, Bahri Community Pharmacy
Suppliers	Nile Medical Supplies, Khartoum Pharma Imports, Gulf MedTrade (USD)
Banks & safes	2 bank journals (SDG/USD) + 2 cash safes (SDG/USD)
Purchases	3 purchase orders received (incl. one in USD); 1 vendor bill
Sales	2 delivered & invoiced orders (SDG) + 1 draft quotation

12. Administration & Maintenance

12.1 Users and access

Create users under **Settings** → **Users**. Assign application access rights so each role sees only the menus it needs (e.g. sales staff vs accountants).

12.2 Backups

Back up the database regularly. The deployment includes backup scripts; on the demo you can also back up from **Database Manager**. Keep off-site copies before upgrades.

12.3 Rebuilding the demo

The demo database can be recreated from scratch using the project's seed script (**scripts/seed_medsupply.py**). See the project README and the **erp-medsupply-demo** runbook for the exact commands.

IMPORTANT — Production hardening

Before real use: change all default passwords, disable the database manager (`list_db = False`), enable HTTPS, and restrict the database filter.

13. Glossary

Term	Meaning
SKU / Internal Reference	Your unique product code.
Lot / Batch	A group of units received together, tracked as one.
FEFO	First Expiry First Out — issue soonest-to-expire stock first.
FIFO	First In First Out — valuation method using oldest cost first.
Putaway rule	Automatic routing of received goods to a specific location.
Journal	A book of account entries (Sales, Purchase, Bank, Cash, Misc).
RFQ	Request for Quotation — a draft purchase order.
Reconciliation	Matching bank-statement lines to invoices and payments.
Base currency	The currency the accounts are kept in (SDG).